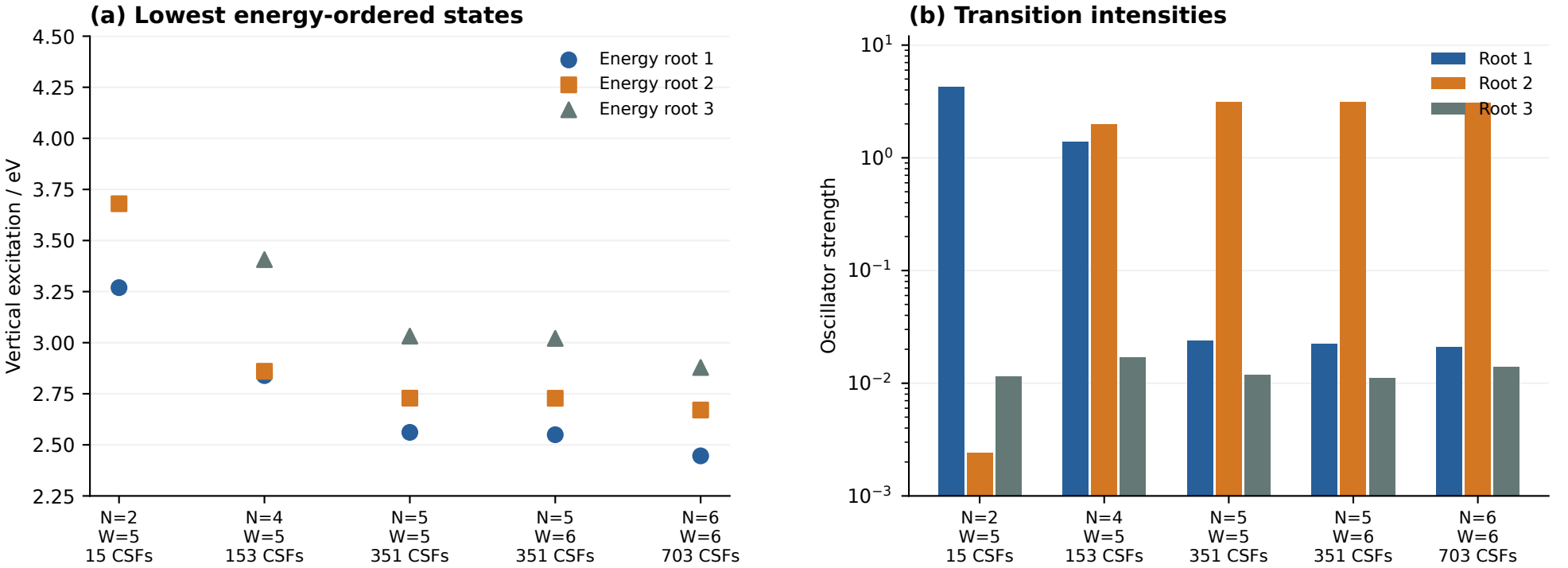
All-trans β-carotene: configuration expansion restores weak S₁ below intense S₂
(a) Lowest energy-ordered states
Vertical excitation / eV
4.50
4.25
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
Energy root 1
Energy root 2
Energy root 3
N=2 W=5 15 CSFs
N=4 W=5 153 CSFs
N=5 W=5 351 CSFs
N=5 W=6 351 CSFs
N=6 W=6 703 CSFs
(b) Transition intensities
Oscillator strength
10¹
10⁰
10⁻¹
10⁻²
10⁻³
Root 1
Root 2
Root 3
N=2 W=5 15 CSFs
N=4 W=5 153 CSFs
N=5 W=5 351 CSFs
N=5 W=6 351 CSFs
N=6 W=6 703 CSFs